

PROJECT90 x Calgary Sensor Lab

Transtibial Prosthetic Pressure Sensor Sock

May 2026 Progress Update

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A Low-Cost, Wireless Prosthetic Sensor Array to Expedite Prosthetic Fitment for
Below-the-Knee Amputations

Painless Prosthetics is a Painful Process

Creating a comfortable prosthetic socket is one of the most clinically important — and technically difficult — challenges in prosthetics.

Today, getting the right fit means months of appointments where results depend on subjective data from the patient. Factors like **mental stress** and **phantom pain** make it challenging for the patient and prosthetist to get fast, reliable results.

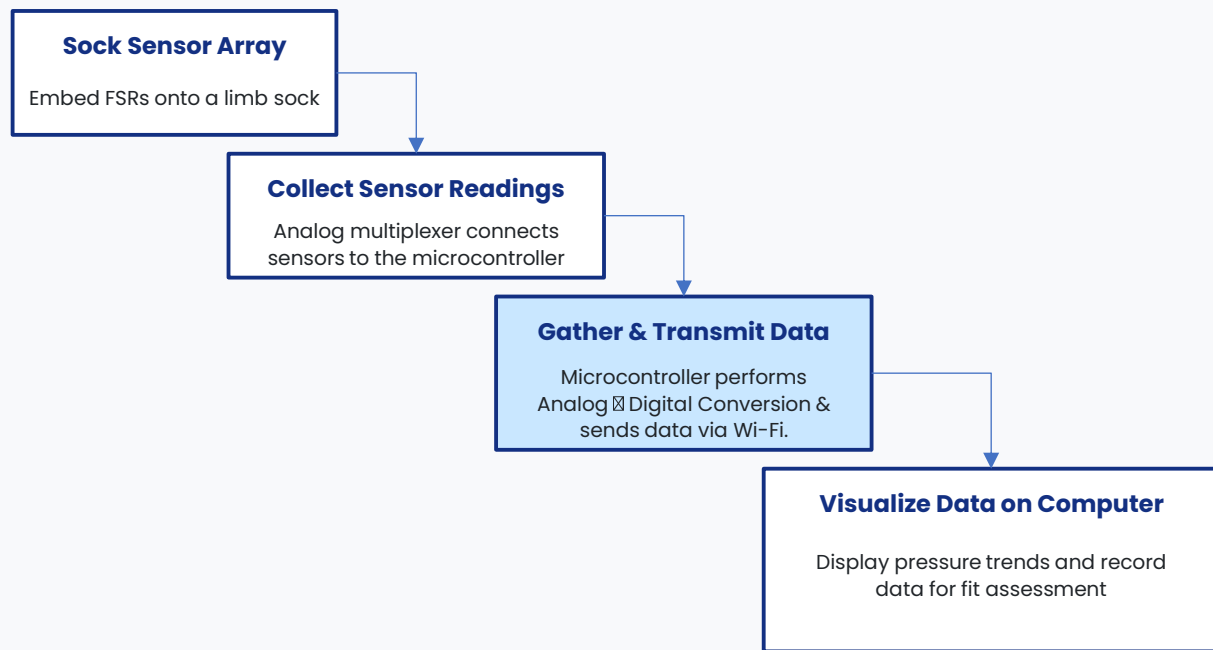
- 01 Pressure distribution at the residual limb–socket interface directly governs comfort, tissue health, and gait quality.** Poor weight distribution can cause pressure sores, skin irritation, and constant irritation which harms liveliness.
- 02 Objective, spatially mapped pressure data could transform the fitting workflow** — enabling prosthetists to quantify load distribution, compare socket iterations, and detect problematic hotspots means more time designing proactively.
- 03 No widely adopted, low-cost, wearable pressure-mapping solution currently exists** for routine clinical use. Existing approaches are a mix of high-cost, unintuitive design, and low accuracy, which create barriers for adoption.

"The goal: a thin, conformable, wireless pressure-mapping sock that makes objective fit data accessible."

A Pipeline of The Pressure Sensor Sock

Here is how we plan to convert a patient's pressure readings into key metrics for clinics

Our sensor array embeds force sensitive resistors onto a standard limb sock. They are routed onto a 16-channel analog multiplexer for data management before connecting to a microcontroller that converts the analog voltage levels into a digital value that is transmitted over Wi-Fi. A computer running the sensor program visualizes this data for the prosthetist. The entire sensor array is powered off a rechargeable battery and wireless connectivity prevents issues with tangled wires and limited mobility in the clinic.

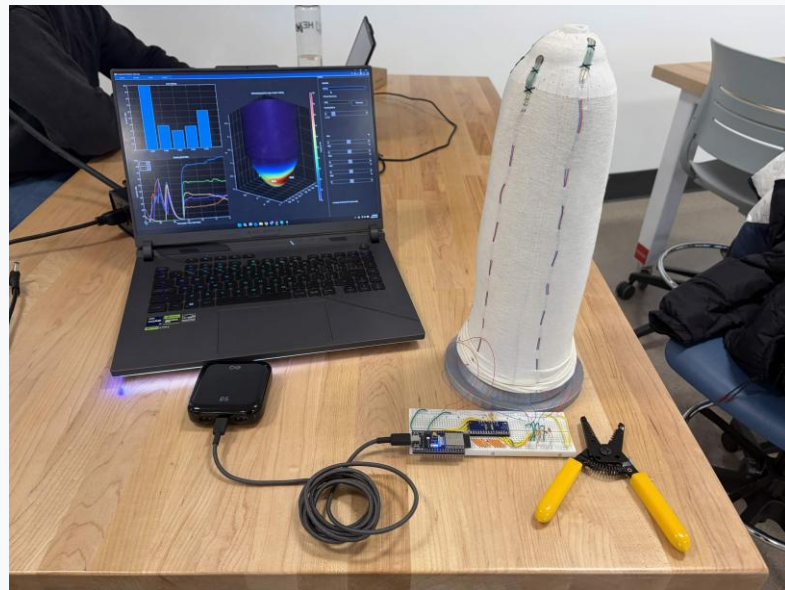


An Overview of Where we are at Right Now

This breadboard, ESP32, and messy wire arrangement is our proof of concept leading into the summer

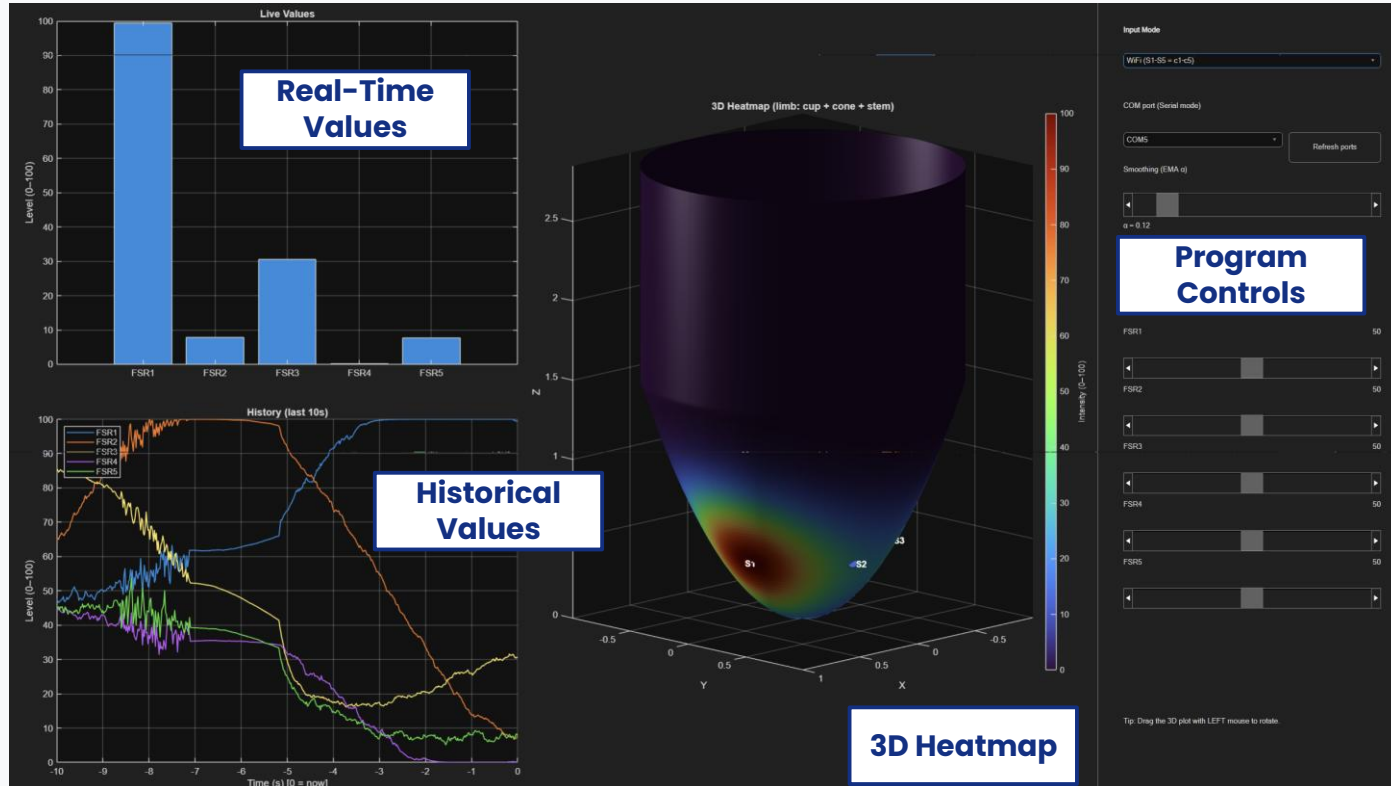
- **Sensors:** 4 x FSR-400 sensors soldered to 30-gauge wire that's been hand-stitched onto a limb sock.
- **Electronics:** ESP32 microcontroller + CD74HC4067 16:1 analog multiplexer on a breadboard with resistors for the voltage divider. Power bank powers everything via USB-C connection.
- **Limb model:** A 3D-printed lower-limb CAD model is used as a test fixture, enabling repeatable sensor placement and load application before human trials.
- **Data Visualization:** We've developed a simple program on the ESP32 to read and transmit sensor readings. The computer runs a MATLAB script to show real-time and historical values as 2D Graphs as well as a 3D real-time heatmap.

Current State: A functional four-sensor proof of concept with wireless connectivity and live data visualization.



MATLAB Simulation Overview

This MATLAB script allows us to rapidly prototype new designs and provides a baseline before we create a custom program.



Assessment of Progress and Challenges

A working demo — with known limitations that define the next phase

WHAT WORKS

- Reliable Wireless Data Pipeline
- Real-time 3D heatmap visualization operational
- Embedded FSRs onto sock with 30-gauge wire for connections and thread for positioning
- Battery-powered sensor array for mobility
- 3D-modelled lower limb for realistic testing.

OPEN CHALLENGES

- **Conductive thread sewing:** 30-gauge wire does limit sock flexibility. Conductive thread would be ideal, but it is prone to snapping under tension and connects poorly to sensors.
- **Analog noise:** Signal noise can result from PCB characteristics and battery-powered sensors.
- **Sensor pressure range:** Off-the-shelf FSRs have not yet been characterized across the local interface pressures expected in use. The prototype still requires controlled load testing.
- **Connector strategy:** Working with 30-gauge wire that's been stitched and soldered is not ideal in clinical conditions. Creating a connector will add structural integrity and clean up wiring.

These issues help us understand what the next task is while working towards our eventual completion

Other Solutions Show Clinical Demand and Areas of Improvement

Recent research supports the feasibility of wearable pressure sensing and highlights the remaining engineering challenges

Recently, a team from Purdue University designed a wearable pressure sensor array with a custom sensor designed to withstand higher pressures. Other solutions like an optical pressure sensor for greater sensitivity have also been attempted.

The research highlights practical requirements for a wearable pressure-mapping system:

- **Wide pressure range** that can withstand an adult's body weight
- **Mechanical Durability** to prevent skewed results under load cycling
- **Comfortable Design** to minimize interference between the limb and socket
- **High Resolution** by maximizing sensor coverage & density
- **Intuitive Data Collection** with a wire-free solution and informative simulations

Traditional FSR-style sensors have raised concerns about accurate readings at higher pressures and the presence of drift over their lifetime. Newer commercial and custom sensors may reduce these limitations, but each option still requires testing in the intended setup.

This project is a direct test as to whether commercial FSRs are now good enough for a practical prosthetic pressure-mapping system.

Project Roadmap

Our team has identified a mix of upcoming improvements and long-term goals

NEAR-TERM PRIORITIES

- 01 **Characterize sensor pressure range** | determine whether commercial FSRs are adequate for human loads
- 02 **Quantify drift, repeatability, and noise** | prevent skewed data using software & determine electromechanical limitations
- 03 **Refine power and signal integrity** | investigate analog noise on battery power & clean up analog pipeline
- 04 **Move from breadboard to PCB** | integrate ESP32, 16:1 MUX, & battery into a smaller footprint. Design sensor connection interface.
- 06 **Expand and refine sensor placement** | optimize array geometry for clinically meaningful pressure mapping coverage

LONGER-TERM / MOONSHOTS

- 01 **Pursue calibrated, quantitative pressure outputs** with reduced drift – moving from relative to absolute pressure measurement
- 02 **Develop a Flexible PCB architecture** for more pressure, larger resolution, and improved flexibility
- 03 **Scale up Architecture for Commercial Production** suitable for wider clinical use
- 04 **Establish Patient Privacy Expectations** for monitoring and storing health data in a legal manner

BUILT TO GROW — LET'S BUILD IT TOGETHER

A working prototype with a clear path forward — and room for collaboration

Our pressure-sensing sock is a working wireless prototype with stitched sensors, live data collection, and 3D visualization. The engineering foundation is in place. What comes next requires YOU.

CLINICAL & BIOMECHANICAL

Guidance on clinically meaningful pressure thresholds, loading scenarios, and socket fit assessment protocols.

SENSOR & MATERIALS

Input on FSR sensor selection, conductive textile integration, and durability under mechanical loading.

SYSTEMS & EMBEDDED

PCB design, enclosure development, sensor connectors, and signal integrity refinement.

Low-cost, wearable, wireless pressure sensing for prosthetics is possible

Let's make it happen